SEMANTIC ART RETRIEVAL WITH FINE-TUNED VISION-LANGUAGE MODELS

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1 PROJECT SUMMARY

Our project investigates text-to-image retrieval for fine art using the SemArt dataset. Given a natural-language description of an artwork, the model will return a ranked list of paintings based on semantic similarity. We will use pretrained CLIP as our baseline and fine-tune it on SemArt image-description pairs to study whether domain-specific adaptation improves retrieval and generalization to unseen artists.

We will also examine how artwork metadata such as artist, style, period, and technique affects retrieval, including whether it provides useful contextual information or creates dataset-specific shortcuts. The model input will be a textual artwork description, optionally combined with metadata, and the output will be a ranked list of paintings.

This problem is challenging because general-purpose vision-language models are trained on broad image-text data, while fine-art descriptions often contain specialized terminology, stylistic references, and historical context that may not align well with those general representations. It is also interesting because successful retrieval requires the model to connect subjective, domain-specific language with subtle visual features rather than relying only on obvious object categories. Our goal is to determine whether adapting CLIP to this domain improves visual-semantic retrieval in a way that generalizes beyond the artists seen during training.

2 RELATED WORK

Garcia & Vogiatzis (2018) introduced SemArt, a multimodal dataset containing fine art images and corresponding attributes, including a textual artistic comment of the painting. Along with this dataset, they introduced *Text2Art*, in which a model would retrieve relevant paintings according to an artistic text, and vice versa. The image and text are encoded into a common semantic space to achieve this retrieval. This work provides both the direct retrieval baseline of our project as well as the primary dataset for our work. For the pretrained baseline model, we utilize CLIP from Radford et al., which learns image and text representations from a large set of general image-text pairs. It offers a useful starting point for our project and a point of comparison after fine tuning on paintings to test the extent to which the art-specific training helps with retrieval.

Conde & Turgutlu (2021) introduced CLIP-Art, which adapts CLIP-style contrastive learning to the fine-art domain using artwork images paired with free-form descriptions or curated artistic attributes. Their method learns joint visual-textual representations and evaluates them on both instance retrieval and fine-grained artwork attribute recognition using the iMet dataset. This work is closely related to ours because it demonstrates that contrastive vision-language pretraining can be successfully specialized for artwork rather than only general web imagery. However, while CLIP-Art focuses primarily on artwork classification and retrieval within iMet, our project applies domain-specific fine-tuning to SemArt and further investigates whether retrieval improvements generalize to artists unseen during training and how additional artwork metadata affects that generalization.

Garcia et al. (2019) built on SemArt by proposing context-aware embeddings that added artwork metadata, such as author, type, timeframe, and school, to assess whether additional context would improve art analysis. They encoded context through a multi-task learning model, where a ResNet50 model learned how the context features related by predicting multiple attributes at the same time. They also used a knowledge graph model, where paintings are nodes that connect to their attributes, node2vec converts the graph into embeddings, and ResNet50 is trained to match them. Their work concluded that models with context outperformed fine-tuned CNNs by up to 7.3% in classification.

This work closely relates to ours as it shows that metadata can improve performance on SemArt. However, this evaluation only uses SemArt's random split, so test paintings may have been created by artists that the model sees during training. In our project, we start with a pretrained CLIP model and assess whether metadata context results in shortcuts or true visual-semantic understanding.

Li et al. (2024) introduced CLIPFit, which adapts pretrained vision-language models through a parameter-efficient fine-tuning method that updates only selected bias terms and normalization layers rather than all model parameters. Their work showed the potential of selective fine-tuning to improve downstream task performance while preserving much of the pretrained model. This closely relates to our exploration in differing fine-tuning configurations of CLIP to fine-art retrieval, except we will be applying this to the more specific SemArt dataset. Also, we have an additional factor of varying the textual context through metadata and artistic commentary associated with their respective images to measure performance differences.

Unlike prior work, we specifically investigate whether CLIP adaptation to SemArt produces retrieval gains that generalize to unseen artists, while separately analyzing whether artwork metadata provides useful semantic context or introduces dataset-specific shortcuts.

3 Proposed Method

The starting point of our project will be pretrained CLIP. CLIP contains an image encoder and a text encoder that map images and text into a shared embedding space. For each SemArt painting, the image encoder will produce an image embedding, while the text encoder will encode the corresponding artistic comment or a user-provided text query. At retrieval time, the query embedding will be compared against the embeddings of candidate paintings using cosine similarity, and the paintings will be returned in ranked order. Our primary baseline will be pretrained CLIP evaluated directly on SemArt without any additional training.

During fine-tuning, we will use CLIP's bidirectional contrastive loss. For each mini-batch, we will compute cosine similarity scores between every image embedding and every text embedding. Matching painting-comment pairs will serve as positive pairs, while the remaining image-text combinations in the batch will act as negatives. Cross-entropy losses will be applied in both the image-to-text and text-to-image directions, encouraging matched pairs to receive higher similarity scores than mismatched pairs.

We will first use computationally efficient adaptation strategies, including training only CLIP's projection layers and partial fine-tuning in which only the final blocks of the image and text encoders are unfrozen. If computational resources permit, we will also evaluate full fine-tuning of all CLIP parameters. Our primary question is whether fine-tuning CLIP on art-specific image-description pairs improves painting retrieval while still generalizing to artists not seen during training.We hypothesize that domain-specific fine-tuning will improve retrieval because training on SemArt should better align CLIP's image and text representations with fine-art-specific visual concepts and terminology. However, gains may be smaller on the artist-disjoint split if the model relies on artist-specific correlations rather than more general visual-semantic relationships. We also expect queries that depend heavily on historical or contextual information not directly visible in the artwork to remain more difficult.

We will evaluate retrieval using Recall@1, Recall@5, Recall@10, and mean reciprocal rank on SemArt's held-out data. We will also construct an artist-disjoint split in which artists appearing in the test set do not appear during training. Comparing results on the standard and artist-disjoint evaluations will allow us to determine whether improvements carry over to genuinely unseen artists or primarily reflect patterns learned from artists represented in the training set.

As a secondary experiment, we will conduct a controlled metadata comparison. The primary retrieval task will use artistic comments alone. We will then augment these descriptions with selected metadata such as artist, style, period, or technique and compare the resulting retrieval performance. By evaluating different metadata subsets separately, we can determine which information improves generalizable retrieval and which may act as shortcuts by making individual artworks easier to identify.

In addition to quantitative evaluation, we will qualitatively inspect representative successful and failed retrievals before and after fine-tuning. In particular, we will compare examples in which the query depends primarily on visible content against examples that require historical, stylistic, or contextual knowledge.

4 DATASETS / ENVIRONMENTS

As previously stated, we will focus on text-to-image retrieval using the SemArt dataset. SemArt contains fine-art painting images paired with artwork attributes and textual artistic comments, making it well suited to our task. To facilitate training, we will preprocess the data by combining selected artwork attributes and/or the corresponding artistic comment into a single text field. The metadata included will vary across experiments, allowing us to compare retrieval performance across the different combinations of metadata and comments. For our primary benchmark, we will use SemArt's established training, validation, and test partitions to facilitate comparison with prior work. We will additionally construct an artist-disjoint partition in which artists appearing in the test set do not appear during training. The SemArt dataset contains 21,390 entries of data, providing a dataset for mini-batch training and comparisons across different configurations. The SemArt dataset is publicly available through its GitHub repository and HuggingFace.

As for the environment, we will conduct training, validation, and testing utilizing the GPUs available with Georgia Tech PACE. This environment supports PyTorch, which we will use in our CLIP-based model implementations.

5 POTENTIAL RISKS

An issue that may impact our project results is that CLIP's text encoder does not take in more than 77 tokens. This is an important factor to consider since many SemArt descriptions may be longer, which may cause us to lose potentially useful information. To address this, we would need to consider truncating or preprocessing the comments to parse out the useful information. For example, we would consider only keeping the first few sentences or splitting a comment to average their embeddings. However, these approaches may cause difficulties in identifying what is considered useful information and may discard details that matter for retrieval. Furthermore, there may be issues with the model finding shortcuts due to artist names in the SemArt comments, which may directly name the artist or related people. To address this, we would need to assess how common it is and whether these names should be masked in the SemArt comments due to significant impact.

SemArt's dataset also contains approximately 21,300 paintings, with about 2,000 reserved for validation and testing. Radford et al. notes that CLIP was initially trained on 400 million image-text pairs. Although this should be enough for lighter fine-tuning, it may present other issues if we fully fine-tune all the weights in the network, as it risks overfitting and losing CLIP's general knowledge. To reduce this risk, we will consider modifying the hyperparameters (e.g. using a lower learning rate) and stop early on the validation set. Furthermore, there may be difficulties in terms of compute limits. We have access to GPUs through PACE at Georgia Tech, but full fine-tuning requires much more GPU memory. If compute limits prevent us from fully fine-tuning our approach, we will likely use a smaller CLIP model or focus on other methods for parameters.

6 CONCLUSION

This project investigates the performance and generalization of text-to-image retrieval of fine-art paintings given a text query, utilizing CLIP-based models and the SemArt dataset. We will compare how retrieval performance between different strategies of fine-tuning and varied artwork metadata associated with images, employing Recall@K and mean reciprocal rank metrics on SemArt's held-out splits.

We will also evaluate the ability of different configurations to generalize paintings made by artists unseen during training. The findings of this project may provide insight into which configurations of pretrained vision-language models may most benefit fine-art retrieval tasks, as well as determining the most effective types of metadata.

7 NEXT STEPS

Next steps here!

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